

CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ

# Chapter 12: Surface Areas and Volumes

Previous year board questions, topic-wise: 5 questions

The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

## 1 Surface area

- Assertion (A): Two identical solid hemispheres joined along their flat faces form a sphere.  
Reason (R): The total surface area of a sphere of radius  $r$  is  $3\pi r^2$ . Choose: (a) both true and R explains A (b) both true but R does not explain A (c) A true, R false (d) A false, R true

CBSE 2025, 1 mark

- A solid sphere is cut into two hemispheres. Find the ratio of the surface area of the sphere to the total surface area of the two hemispheres together. (a) 1 : 1 (b) 1 : 4 (c) 2 : 3 (d) 3 : 2

CBSE 2024, 1 mark

## 2 Volume

- A room has the shape of a cylinder with a hemispherical dome on top. The radius of the dome is half the height of the cylindrical part, and the room holds  $\frac{1408}{21}$  m<sup>3</sup> of air. Find the height of the cylindrical part. Take  $\pi = \frac{22}{7}$ . CBSE 2025, 3 marks
- The largest possible right circular cone is cut from a solid cube of edge 2 cm. Find its volume.  
(a)  $\frac{4\pi}{3}$  cu cm (b)  $\frac{5\pi}{3}$  cu cm (c)  $\frac{8\pi}{3}$  cu cm (d)  $\frac{2\pi}{3}$  cu cm CBSE 2024, 1 mark
- A hollow cylinder is 14 cm long. Its outer radius is 1 cm more than its inner radius, and the metal in it has volume 176 cm<sup>3</sup>. Find the two radii. CBSE 2024, 3 marks

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